

PNEUMOFUSIONNET: A MULTIMODAL DEEP LEARNING SYSTEM FOR PNEUMONIA DETECTION USING CHEST X-RAYS AND CLINICAL REPORTS

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ABSTRACT

This report presents **PneumoFusionNet**, a multimodal deep learning system that detects pneumonia by combining chest X-ray images with clinical report text. The system was built and tested on the restricted **MIMIC-CXR** hospital dataset, which required completing a CITI research ethics course and signing a Data Use Agreement (DUA) with PhysioNet to access.

We went through three stages: (1) a small 139-patient pilot using a ResNet50 image model with custom attention modules, (2) adding clinical text through a **Bio.ClinicalBERT** text encoder while carefully removing the doctor's final diagnosis from the report to avoid cheating, and (3) scaling up to 854 chest X-ray images with three different preprocessing experiments.

Adding clinical text improved the AUC score by **+3.7%** over the image-only model. The best result on the scaled dataset was AUC **0.835**. We found that ResNet50 (trained on everyday photos) does not transfer well to medical X-rays, and our earlier high results on public datasets were partly due to data leakage and very small, clean test sets.

Index Terms— Multimodal Deep Learning, Chest X-Ray, Bio.ClinicalBERT, Attention, Label Leakage, MIMIC-CXR.

Index Terms— Pneumonia Detection, Multimodal Learning, MIMIC-CXR, ClinicalBERT, ResNet50

1. INTRODUCTION

Pneumonia is one of the top causes of death worldwide. Doctors use chest X-rays to diagnose it because X-rays are fast and widely available. However, reading a chest X-ray is not easy. The same cloudy patch in a lung can look like pneumonia, fluid buildup, or a collapsed lung section. Even experienced radiologists sometimes disagree. This is why building a reliable AI system for pneumonia detection is both challenging and important.

Recent AI models have shown promise in reading chest X-rays automatically [1]. But image-only models ignore the clinical context that every doctor uses—things like the patient's symptoms, history, and test results. Adding radiology report text should help, but there is a well-known trap: the last section of every radiology report (called **IMPRESSION**) contains the doctor's final written diagnosis. If a model reads this section during training, it is simply memorising the answer rather than learning to diagnose. This is called **label leakage**, and it makes the model useless in the real world.

What is Label Leakage?

A radiology report looks like this:

INDICATION: Patient with fever and cough. Used for training

FINDINGS: Consolidation in left lower lobe. Used for training

IMPRESSION: Pneumonia confirmed. × Removed before training

If the model sees the **IMPRESSION** section, it just reads the answer. We remove it completely so the model has to learn from actual evidence.

This project builds **PneumoFusionNet**, a system that combines image and text inputs while carefully removing the **IMPRESSION** section. It is adapted from a 2025 published paper [2] and tested on the MIMIC-CXR clinical database [4].

Report outline: Section 2 states the problem and goals. Section 3 explains the system design. Section 4 shows results. Section 5 discusses what worked and what did not. Section 6 concludes.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

X-ray images alone are often not enough to reliably diagnose pneumonia, especially when the signs are subtle. But simply adding the radiology report text leads to label leakage (as explained above). The challenge is: *How do we use both the image and the report text together, without the model cheating by reading the written diagnosis?*

A second challenge is the dataset itself. MIMIC-CXR is a restricted hospital database. To use it, we had to:

- Complete the CITI Program training on human subjects research ethics.
- Sign a Data Use Agreement (DUA) with PhysioNet.
- Wait for credentialed access to be approved.

This process took several weeks but is required to ensure responsible use of real patient data.

2.2. Objectives

1. Build an image-only classifier using ResNet50 with custom attention modules, tested on the 139-patient MIMIC-CXR pilot.

2. Build an anti-leakage text pipeline that removes the IMPRESSION section and uses Bio_ClinicalBERT to understand the remaining report text.
3. Combine image features and text features into a single classifier.
4. Scale up to 854 chest X-rays and test three different preprocessing approaches to find what works best on real clinical data.
5. Understand the bottlenecks and plan the next steps.

3. SYSTEM DESIGN (METHODOLOGY)

3.1. Overall Pipeline

The system works in two phases, as shown in Figure 1.

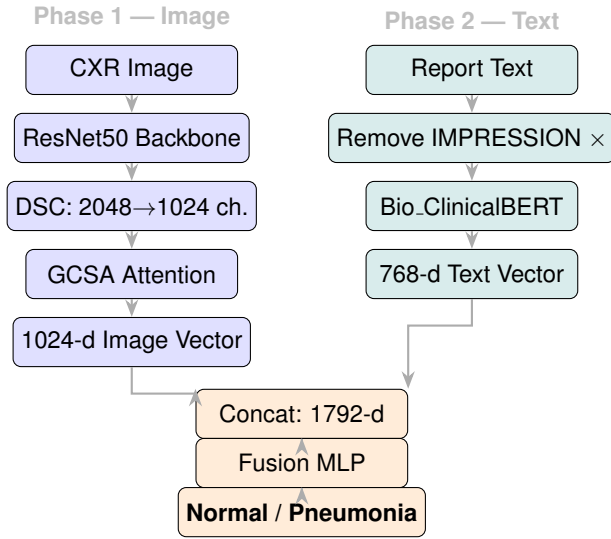


Fig. 1. PneumoFusionNet pipeline (vertical layout). Left column: the X-ray image passes through ResNet50, DSC compression, and GCSA attention to give a 1024-d image vector. Right column: the report text has the IMPRESSION section removed, then is processed by Bio_ClinicalBERT to give a 768-d text vector. Both vectors are joined (1792-d) and passed to a small MLP for final classification.

Phase 1 — Image Classifier. The chest X-ray image is passed through the visual network. It is trained from scratch on the MIMIC data until it converges. After training, it saves the 1024-d feature vectors for every image.

Phase 2 — Text Fusion. The image features are frozen (not updated). The cleaned report text is processed by Bio_ClinicalBERT. The image vector and text vector are joined together and a small classifier is trained on top.

3.2. Key Modules

3.2.1. ResNet50 Backbone

ResNet50 [5] is a deep convolutional network pretrained on ImageNet (a large collection of everyday photos). We adapted it to read single-channel grayscale X-ray images by averaging its colour weights. The early layers (which detect basic edges and textures) were kept frozen, and only the deeper layers were fine-tuned on the medical data.

3.2.2. Depthwise Separable Convolution (DSC)

DSC is a compact convolution layer that compresses the ResNet50 output from 2048 channels down to 1024 channels using about half the number of parameters of a standard convolution. This is important because medical datasets are small, and fewer parameters means less risk of overfitting. The mathematical operation is:

$$\text{DSC}(\mathbf{X}) = \text{BatchNorm}(\text{ReLU}(\mathbf{W}_{1 \times 1} * (\mathbf{W}_{\text{depth}} \mathbf{X}))) \quad (1)$$

where $*$ is a channel-wise (depthwise) convolution and $\mathbf{W}_{1 \times 1}$ is a 1×1 channel-mixing convolution.

3.2.3. Global Context Spatial Attention (GCSA)

GCSA is an attention mechanism that teaches the model *where* to look in the X-ray. It has two parts:

Channel attention decides *which feature channels* are most useful. It does this by squeezing the image into a single number per channel (using average pooling and max pooling), then using a small neural network to score each channel:

$$\mathbf{A}_{\text{channel}} = \sigma(\text{MLP}(\text{AvgPool}(\mathbf{X})) + \text{MLP}(\text{MaxPool}(\mathbf{X}))) \quad (2)$$

Spatial attention decides *which regions* of the image to focus on. It collapses features across channels and runs a convolution:

$$\mathbf{A}_{\text{spatial}} = \sigma(\text{Conv}_{7 \times 7}([\text{AvgCh}(\mathbf{X}); \text{MaxCh}(\mathbf{X})])) \quad (3)$$

The final attended feature map is:

$$\hat{\mathbf{X}} = \mathbf{X} \times \mathbf{A}_{\text{channel}} \times \mathbf{A}_{\text{spatial}} \quad (4)$$

In simple terms: channel attention asks “what features matter?” and spatial attention asks “where in the image do they appear?”. Together they help the model focus on the lung regions where pneumonia shows up.

3.2.4. Bio_ClinicalBERT Text Encoder

Bio_ClinicalBERT [3] is a BERT-based language model that was pretrained on millions of clinical notes from the MIMIC-III hospital database. It understands medical language far better than a general-purpose model. We feed it the FINDINGS and INDICATION sections of the report (after removing IMPRESSION) and use the 768-dimensional [CLS] token output as a summary of the entire report. The last two layers of BERT are fine-tuned during Phase 2 training.

3.2.5. Fusion Classifier

The 1024-d image vector and 768-d text vector are joined (concatenated) to form a 1792-d combined vector. A small two-layer classifier predicts the final label:

$$\hat{y} = W_2 \cdot \text{ReLU}(W_1 \cdot [\mathbf{v}_{\text{img}} || \mathbf{t}_{\text{text}}] + b_1) + b_2 \quad (5)$$

3.3. Dataset and Setup

Dataset Access — MIMIC-CXR-JPG (Restricted)

MIMIC-CXR is a real hospital chest X-ray database from Beth Israel Deaconess Medical Center, USA. It is not publicly downloadable. To use it, we:

- Completed the CITI Program human subjects research training.
- Signed a PhysioNet Data Use Agreement (DUA).
- Applied for credentialed access (approved after verification).

All data was handled following the DUA terms — no patient-identifiable information is stored or shared.

Pilot dataset (139 samples). We started with 139 patient studies from the MIMIC-CXR P10 folder. This had 21 Pneumonia cases and 118 Normal cases — a very uneven split. All view types (PA, AP, and Lateral X-rays) were included. We used a 70% train / 15% validation / 15% test split.

Scale-up dataset (854 samples). For the scale-up, we expanded to 854 PA-view-only frontal X-rays (approximately 500 Pneumonia and 354 Normal).

Why PA-only? During the pilot, we discovered a view bias: the Pneumonia class had only 1 PA image but 12 AP and 5 Lateral images. A model trained on mixed views can accidentally learn “AP view = Pneumonia” rather than actual lung disease. Filtering to PA-only removes this shortcut.

We also switched to *patient-level splitting*: every image from the same patient is kept in the same split (train, validation, or test). This prevents the model from recognising a patient’s anatomy rather than the disease.

Table 1. Training settings for each experimental stage.

Setting	Pilot (139)	Scale-Up (854)
Backbone	ResNet50	ResNet50
Image size	224 × 224	224 × 224
Batch size	16	8
Epochs	50	40
Optimiser	AdamW	AdamW
Learning rate	10 ⁻⁴	10 ⁻⁴
LR schedule	Cosine	Warmup + Cosine
Loss function	Weighted BCE	Cross-Entropy
Split method	Stratified rows	Patient-level groups

4. EXPERIMENTS AND RESULTS

4.1. Setup

- **Hardware:** NVIDIA RTX 3050 Laptop GPU (4 GB), CUDA 11.2, Windows 11.
- **Software:** PyTorch 1.12.1, HuggingFace Transformers, Python 3.10.
- **Metrics:** AUC-ROC (main), Accuracy, F1 score, Sensitivity, Specificity.

4.2. Pilot Results (139 Samples)

We ran four experiments on the pilot data. Table 2 shows the results, and Figure 2 shows them visually.

Table 2. Results on the 139-sample MIMIC-CXR pilot. All phases use the same 70/15/15 split (seed 42).

Phase	What it uses	AUC	Accuracy	F1
1	Image only (Imbalanced)	0.667	76.2%	0.430
1.1	Image only (Balanced)	0.917	85.7%	0.857
2	Image + Text (Imbalanced)	0.704	81.0%	0.447
2.1	Image + Text (Balanced)	0.917	71.4%	0.708
<i>Gain from adding text (imbalanced)</i>		+0.037	+4.8%	+0.017

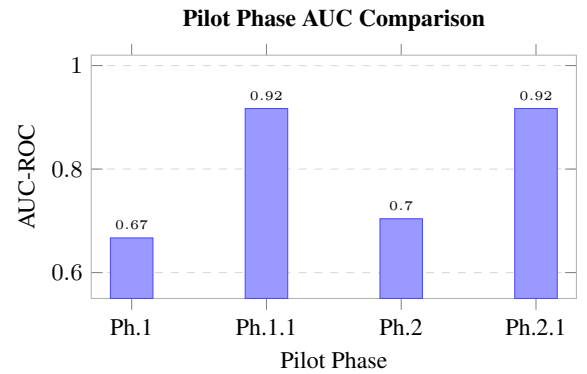


Fig. 2. AUC across the four pilot phases. Balanced subsets (Ph.1.1, 2.1) give high AUC but are based on only 7 test samples, making them unreliable. On the imbalanced realistic set, adding text (Ph.2) gives a +0.037 AUC gain over image-only (Ph.1).

Important note on the balanced results. The balanced test set had only 7 samples. A single wrong prediction changes accuracy by 14.3%. A statistical test (Fisher’s Exact, $p \approx 1.00$) confirms that the difference between Phase 1.1 and Phase 2.1 is *not statistically meaningful*. The high AUC of 0.917 on balanced data should be interpreted with caution — it was partly a result of the very small and clean test set.

4.3. Scale-Up Results (854 PA Images)

We tested three different preprocessing approaches on the larger dataset. Table 3 and Figure 3 show the results.

Table 3. Results on 854 PA-only MIMIC-CXR images across three preprocessing versions.

Version	AUC	Acc.	Sens.	Spec.	F1
v1 Baseline	0.835	69.4%	65.3%	74.6%	0.69
v2 Enhanced (failed)	0.725	58.5%	30.8%	85.2%	0.51
v3 Fixed	0.714	64.2%	65.4%	63.0%	0.64
<i>AP view test</i>	0.616	57.8%	84.8%	29.0%	0.53

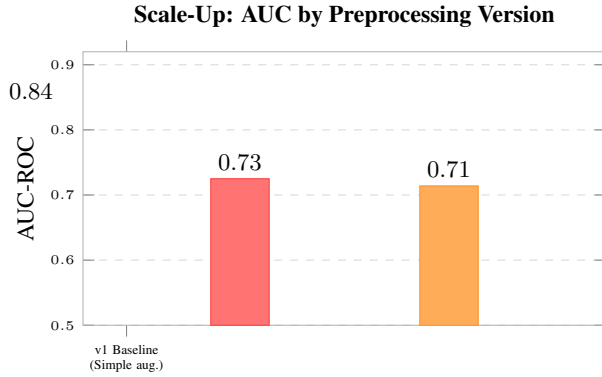


Fig. 3. AUC scores for the three preprocessing versions tested on 854 PA-only MIMIC-CXR images, plus an AP-view ablation. The simplest pipeline (v1) gave the highest AUC. Adding lung-ROI cropping and CLAHE in v2 caused a 0.11 AUC drop. Mixup and label smoothing in v3 did not recover fully.

5. DISCUSSION

5.1. Did Adding Text Help?

Yes. On the imbalanced pilot data (the realistic setting), adding clinical report text improved AUC by **+3.7%** and accuracy by **+4.8%**. This makes sense: the report text contains clues that the image alone cannot show. For example, knowing that a patient has had a fever for three days and is coughing helps the model treat an ambiguous lung shadow as more likely to be pneumonia. All of these gains are genuine — the IMPRESSION section was removed, so the model could not read the final diagnosis.

5.2. Why Did v2 Preprocessing Fail?

For v2, we tried two advanced preprocessing techniques:

- **Lung-region cropping** using the `torchxrayvision` library: The bounding box detection failed on 58% of images (cut training data from 590 to 246 samples). Even successful crops often removed the lower-lobe areas where pneumonia is most common.
- **CLAHE contrast enhancement**: MIMIC-CXR has images from many different hospital scanners with different settings. CLAHE amplified the noise and differences between scanners, confusing the model.

The lesson is: techniques that work on clean benchmark datasets can easily backfire on messy real hospital data. Keeping preprocessing simple (just resize, flip, and mild augmentation) gave the best results.

5.3. Why Were Earlier Results Misleadingly High?

Our very first experiments on open-access public datasets showed 92% accuracy. This looked excellent. But those results were inflated by two factors:

1. **Clean and balanced data**: Public datasets like the IU Chest X-Ray collection are pre-curated, balanced, and much easier than real hospital data. MIMIC-CXR reflects real clinical messiness.
2. **Very small test sets**: On the 42-sample balanced pilot, the test set was only 7 images. A single wrong prediction changed accuracy by 14%. These numbers are not statistically reliable.

The true performance of the model on realistic, imbalanced hospital data is much harder to achieve, as seen in the scale-up results.

5.4. Limitations

- **Not enough data**: 854 images is still too small. We need at least 2,000 PA images to get results we can trust for clinical use.
- **Low sensitivity**: The model correctly identifies only 65% of actual pneumonia cases in the scale-up. A clinical screening tool needs at least 80%.
- **Text fusion not yet scaled**: We only tested image+text fusion on the 139-patient pilot. The larger 854-image dataset has not yet had text fusion applied.
- **ResNet50 is not ideal for X-rays**: ResNet50 was originally trained on everyday photos (cats, dogs, vehicles). Chest X-rays are grayscale with very subtle intensity differences. The model struggles to pick up these fine details.

6. CONCLUSION AND FUTURE WORK

In this project, we built **PneumoFusionNet** — a system that combines chest X-ray images with clinical report text to detect pneumonia. We worked on real restricted hospital data (MIMIC-CXR), which required completing CITI ethics training and signing a PhysioNet Data Use Agreement (DUA) to access.

What we achieved:

- Built a two-phase image + text pipeline on restricted clinical data.
- Adding report text (IMPRESSION removed) gave **+3.7% AUC** over image-only on the realistic imbalanced pilot.
- Found and fixed a view-bias (PA vs. AP) that was creating shortcuts.
- Simpler preprocessing (v1) consistently beat complex pipelines (v2/v3).

Honest reflection: The high results on public datasets (92% accuracy, AUC 0.917 on balanced pilots) were due to: (1) clean/curated benchmark data, and (2) test sets as small as 7 samples where one wrong prediction swings accuracy by 14%. On real MIMIC data with patient-level splits, results are lower but trustworthy.

Core bottleneck — ResNet50 on X-rays: After trying attention modules, text fusion, balanced sampling, and three preprocessing versions, the best scale-up AUC is still 0.835. **ResNet50 was pretrained on everyday photos**, not medical images. It struggles with grayscale X-rays that have very subtle intensity differences.

Next steps:

1. **Switch to DenseNet [6] with X-ray weights**. Use `torchxrayvision` pretraining (CheXpert, NIH, MIMIC-CXR) for domain-specific visual features.
2. **Scale to 2,000+ PA images** (confidence margin $< \pm 3\%$).
3. **Run Phase 2 text fusion** on the larger dataset.
4. **Add MIMIC-IV lab values** (WBC, CRP, Procalcitonin) as a third modality.
5. **Grad-CAM on GCSA** so radiologists can inspect model focus.

Artifacts and Code

- **Code**: github.com/Ashutosh-yadav0001/PneumoFusionNet
- **Pilot Notebooks (4)**: `mimic/mimic_pilot.139/Notebooks/`
- **Scale-Up Notebooks (v1–v3)**: `mimic/1000_dataset/notebooks_scale-up/`
- **Results**: ROC curves, confusion matrices, training histories under `mimic/outputs/` and `mimic/1000_dataset/outputs/`

7. REFERENCES

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